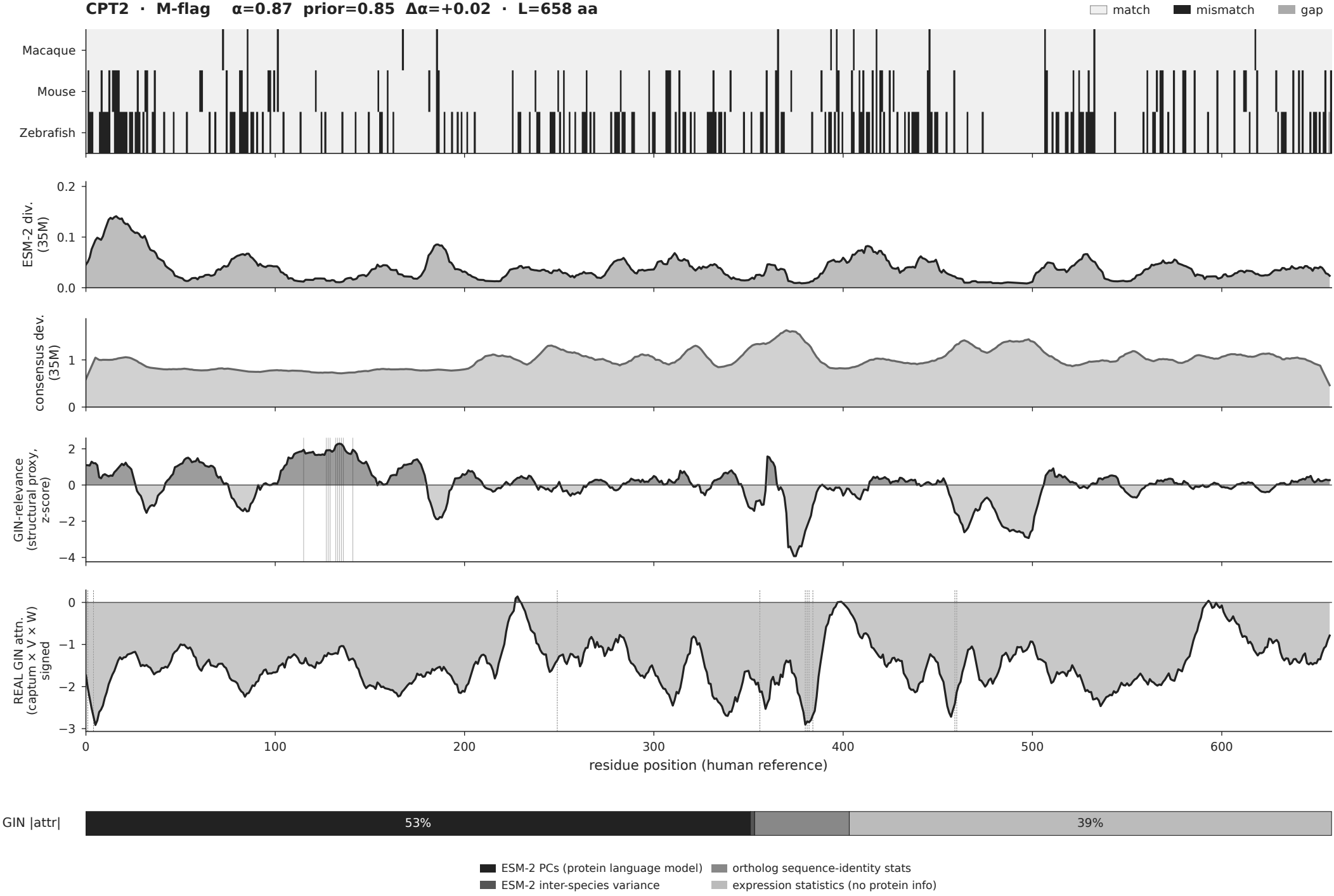

CPT2 · M-flag α=0.87 prior=0.85 Δα=+0.02 · L=658 aa
match
mismatch
gap
Macaque
Mouse
Zebrafish
ESM-2 div. (35M)
0.2
0.1
0.0
consensus dev. (35M)
1
0
GIN-relevance (structural proxy, z-score)
2
0
−2
−4
REAL GIN attn. (captum × V × W) signed
0
−1
−2
−3
0
100
200
300
400
500
600
residue position (human reference)
GIN |attr|
53%
39%
ESM-2 PCs (protein language model)
ortholog sequence-identity stats
ESM-2 inter-species variance
expression statistics (no protein info)